

Who are you?

MBTI-Personality Type Classification

AUTHORS

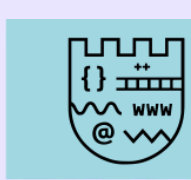
Kontos Christos
Efthymiadis Konstantinos

CONTACT INFO

ckontosb@csd.auth.gr
kefthyme@csd.auth.gr

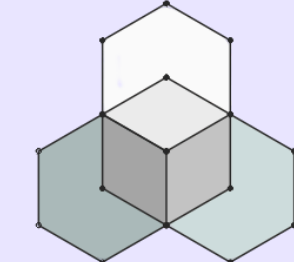


ARISTOTLE
UNIVERSITY
OF THESSALONIKI



SCHOOL OF
INFORMATICS

Data & Web
Science

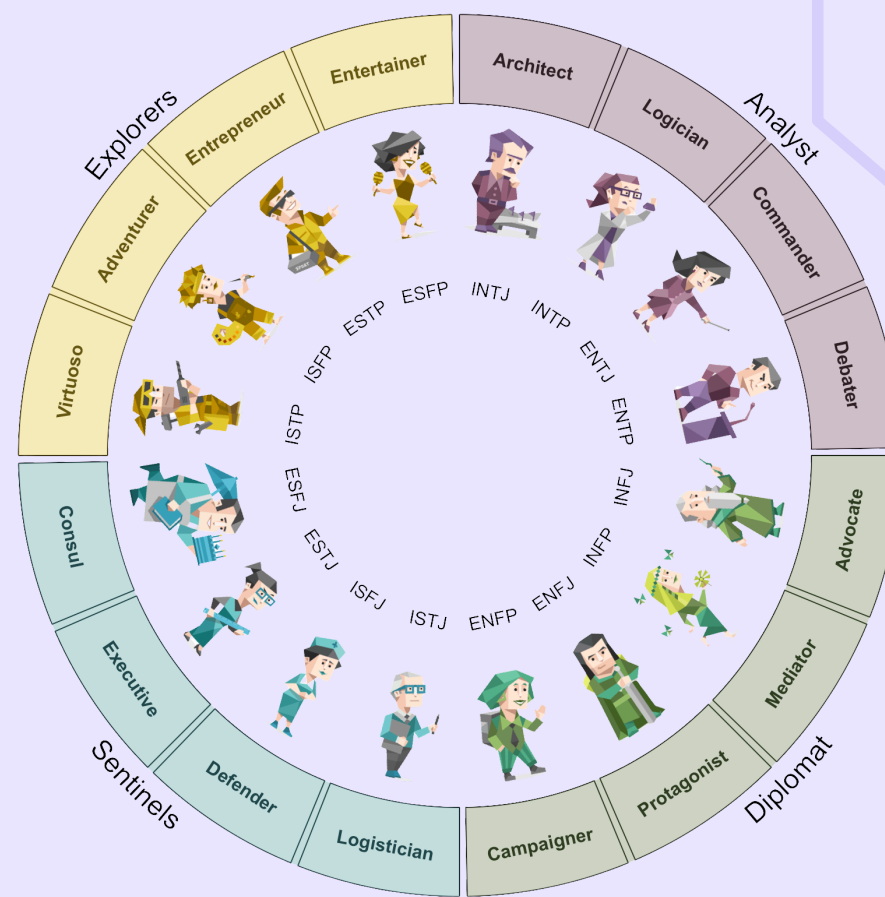


MSc Program



Introduction

- The Myers-Briggs Type Indicator, or **MBTI**, is a personality framework that groups people into 16 personality types based on four dimensions:
 - Extraversion-Introversion
 - Intuition-Sensing
 - Thinking-Feeling
 - Judging-Perceiving
- Since each MBTI type is made up of one choice from each dimension, the prediction task can be split into four binary classification problems: EI, NS, TF, and JP.
- In this work, we use user-level text to predict MBTI personality types and compare classical multi-label machine learning models with transformer ones.



Dataset

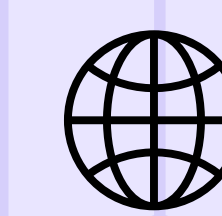


Training Dataset: Kaggle MBTI dataset

- 8600 users
- 50 posts/users



Unit of Analysis: User-level text, not individual posts



Pandora is used for zero-shot prediction



Methodology

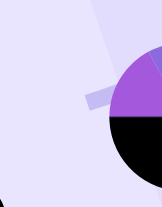
Preprocessing



Text Cleaning: Remove URLs, separators, punctuation noise, MBTI type words



Extract Labels: EI, NS, TF, and JP



Stratified Split:
70% train / 15% validation / 15% test

Models

Classical ML

TF-IDF Vectorization

- Logistic Regression
- LinearSVC
- Classifier Chains
- Label Powerset

Transformer

Chunking

Tokenization

RoBERTa

MentalRoBERTa

ModernBERT

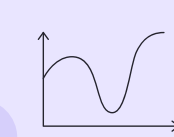
4 Axis Head

E/I N/S T/F J/P

Evaluation



Macro-F1



ROC-AUC



Accuracy



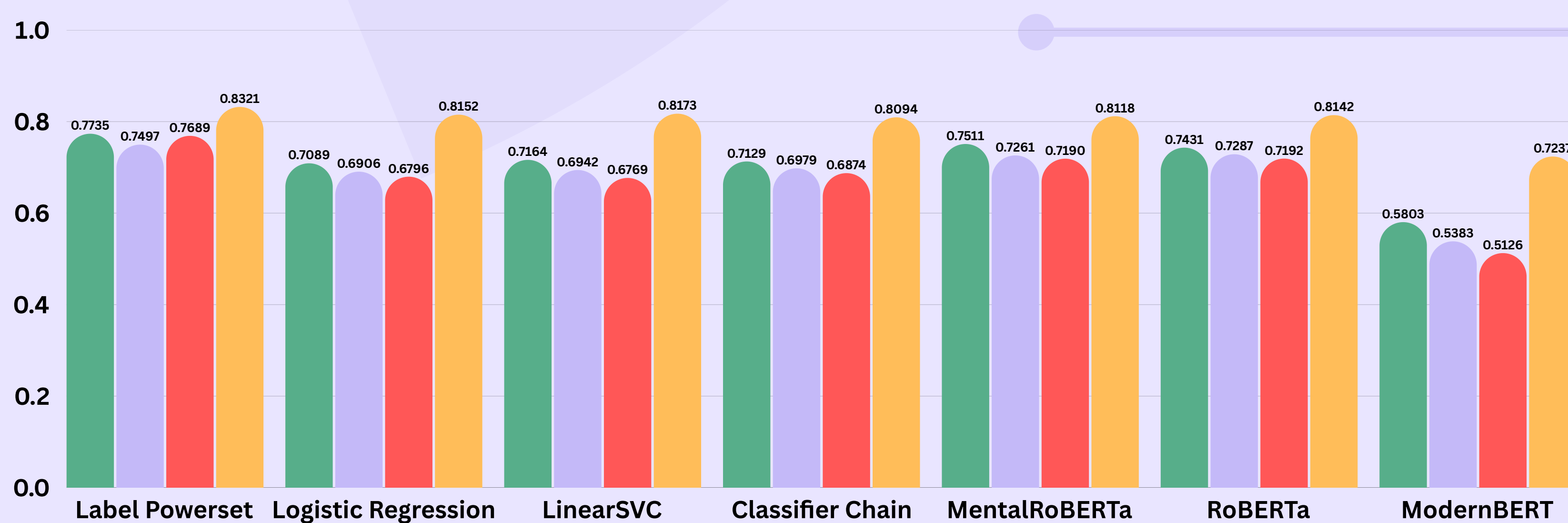
Zero Shot Ocean



Results

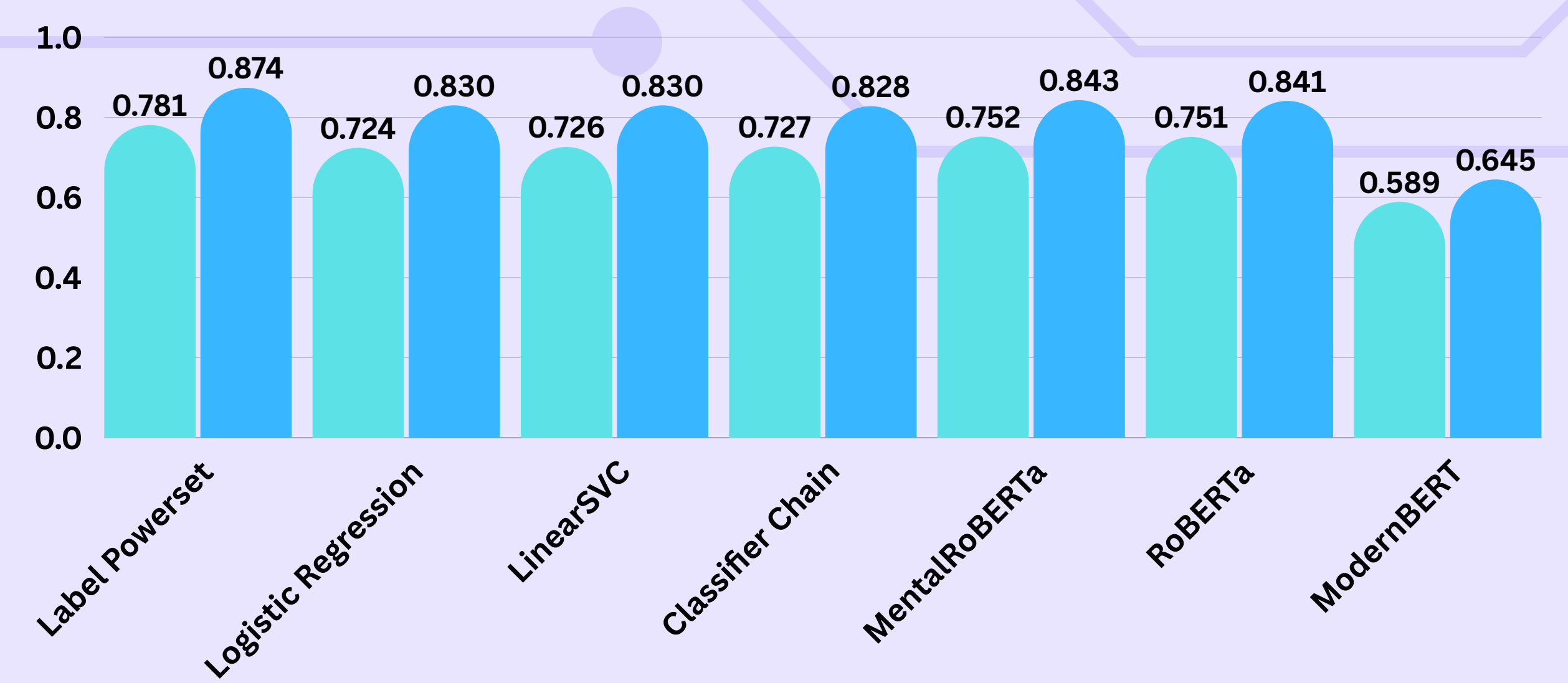
MBTI macro-F1 per axis

E/I J/P N/S T/F



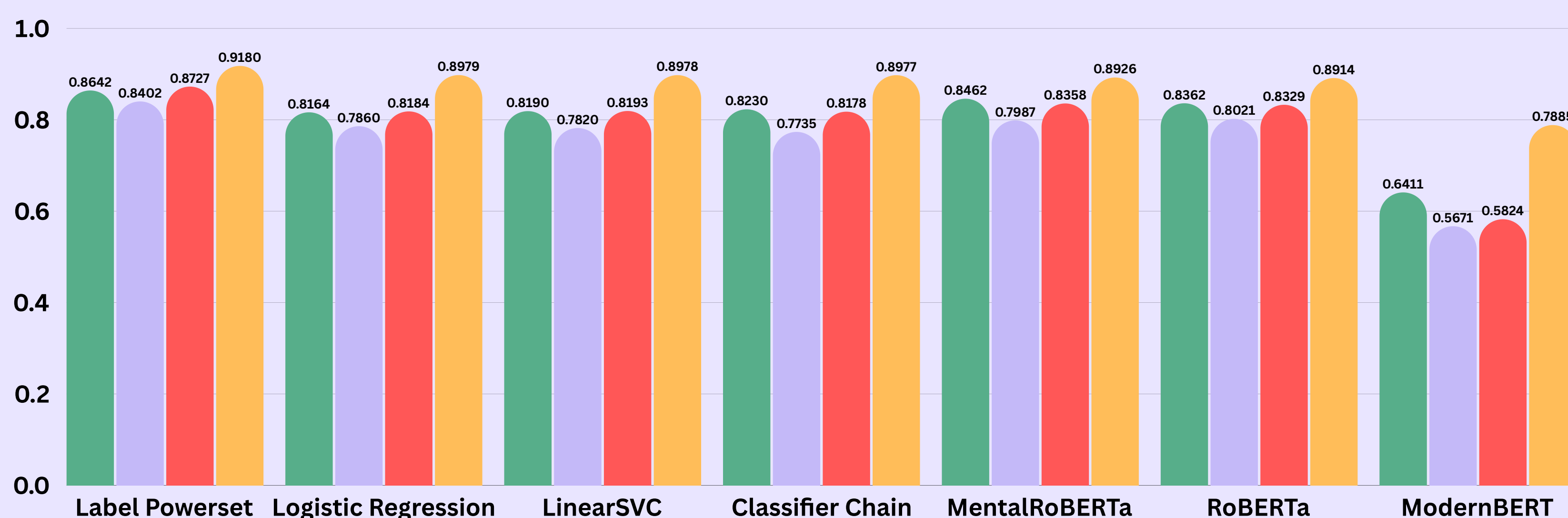
MBTI macro-F1

F1-macro ROC-AUC



MBTI ROC-AUC per axis

E/I J/P N/S T/F



Pandora OCEAN Transfer: median-split ROC-AUC

E/I J/P N/S T/F

